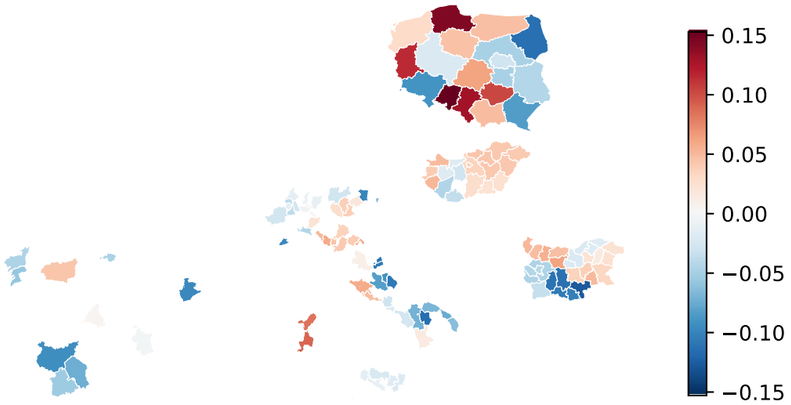
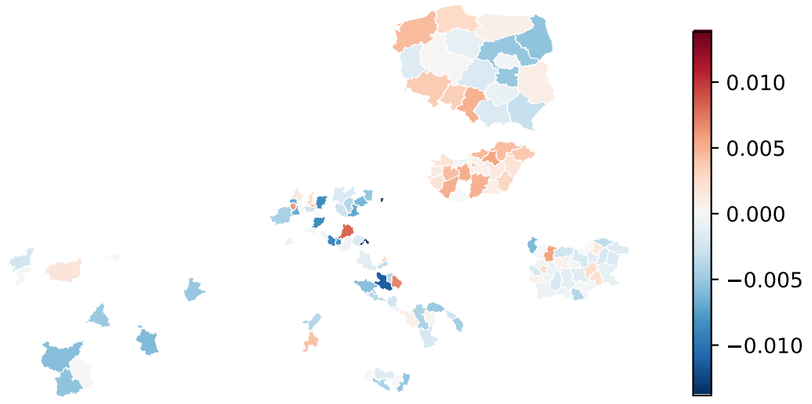
Need line (quantile regression): how each selected variable pushes households towards (red) or away from (blue) Double Risk, regional mean SHAP (CatBoost, pooled five-country model, stage B2)

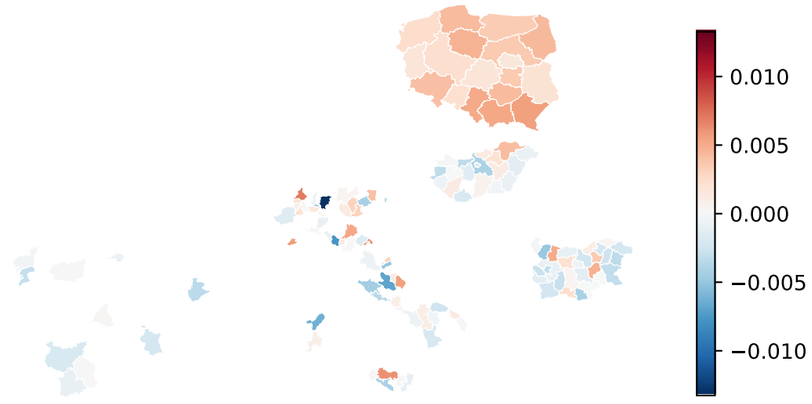
region (NUTS2)



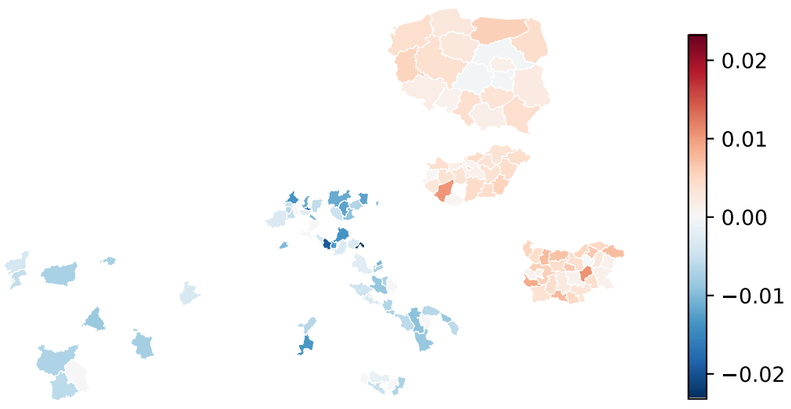
wealth index (house-value stand-in)



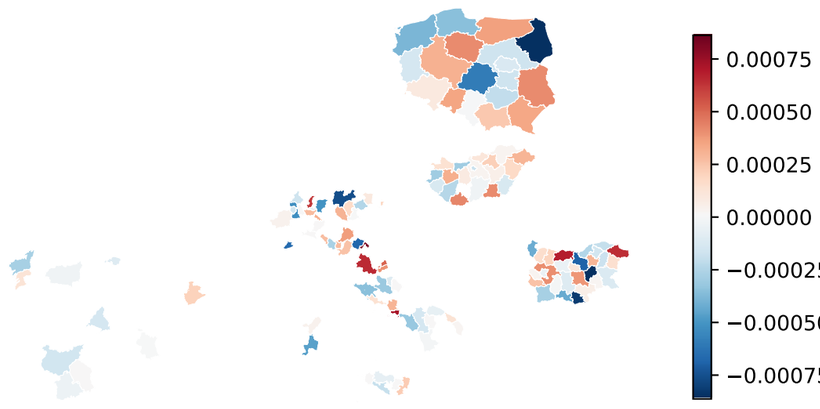
household vulnerability index



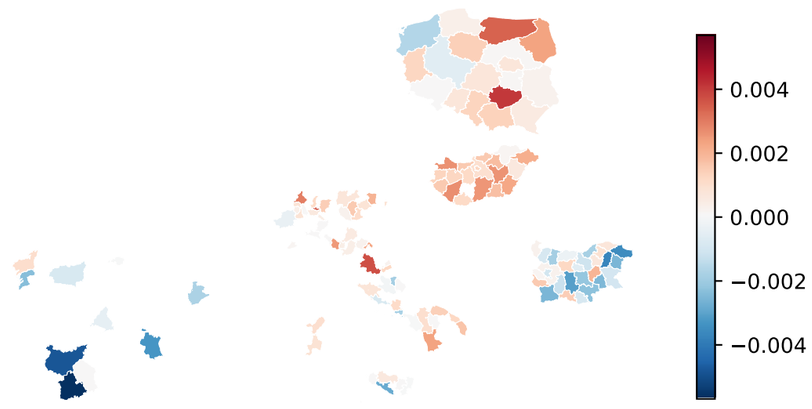
income per adult equivalent (EUR)



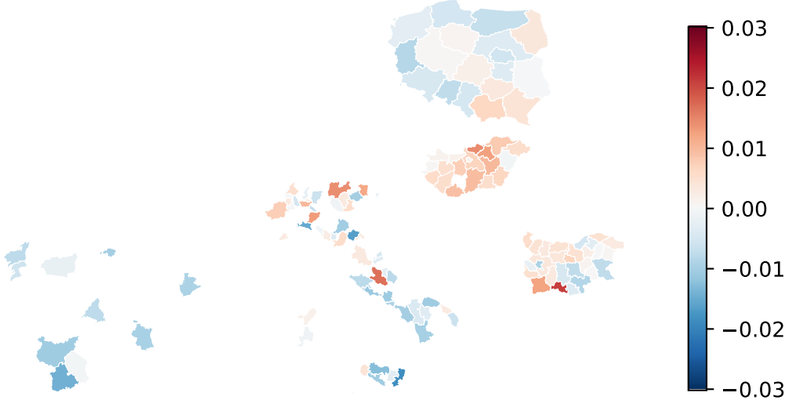
inefficient equipment index



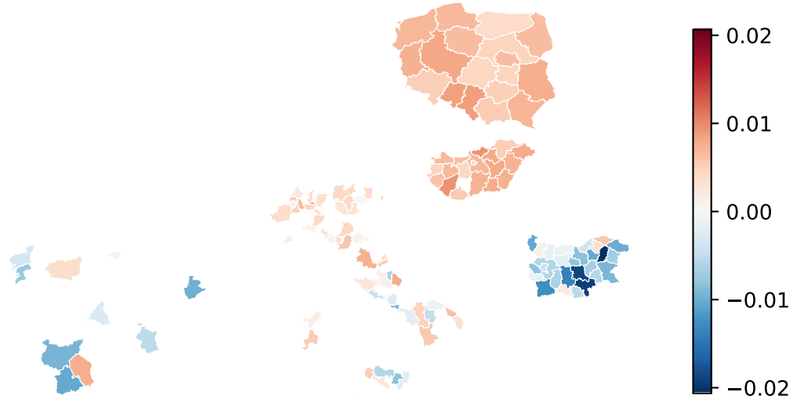
heating system type



heating need per person



cold winter x electric heating



density x urban area

